

# Modern Algebra 1

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Grant Talbert  
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## Problem 1

*Proof.* (a)

$$\begin{aligned}\phi(f_1, \dots, f_n) \phi(\ell_1, \dots, \ell_n) &= (a_1^{f_1} \cdots a_n^{f_n}) (a_1^{\ell_1} \cdots a_n^{\ell_n}) \\ &= a_1^{f_1} a_1^{\ell_1} \cdots a_n^{f_n} a_n^{\ell_n} \\ &= a_1^{f_1 + \ell_1} \cdots a_n^{f_n + \ell_n} \\ &= \phi(f_1 + \ell_1, \dots, f_n + \ell_n).\end{aligned}$$

Additionally, notice that

$$(f_1, \dots, f_n) + (\ell_1, \dots, \ell_n) = (f_1 + \ell_1, \dots, f_n + \ell_n).$$

Therefore,  $\phi$  is a homomorphism.

(b) First we show  $\phi(F) = G$ . Notice that for all  $a_i \in G \setminus \{e\}$ , there exists  $(f_1, \dots, f_n) \in F$  such that  $f_j = 0$  for all  $j \neq i$ , and  $f_i = 1$ . Notice that

$$\phi(0, \dots, 1, \dots, 0) = a_1^0 \cdots a_i^1 \cdots a_n^0 = e \cdots a_i \cdots e = a_i.$$

Therefore, for all  $a_i \in G \setminus \{e\}$ , there exists  $f \in F$  such that  $f \mapsto a_i$ . Furthermore, note that for  $(0, \dots, 0) = e_F \in F$ , and  $(0, \dots, 0) \mapsto e_G$  by properties of homomorphisms. Therefore, for all  $g \in G$ , there exists  $f \in F$  such that  $f \mapsto g$ , and  $\phi$  is surjective. It follows that,  $G \subseteq \phi(F)$ . Finally, note that  $\phi(f) \in G$  for all  $f \in F$ , so  $\phi(F) = G$ .

We will now apply the first isomorphism theorem. Note that

$$F / \ker \phi \cong \phi(F) = G.$$

By properties of isomorphisms and quotient groups, we have

$$|G| = |F / \ker \phi| = \frac{|F|}{|\ker \phi|}.$$

It follows that

$$|G| |\ker \phi| = |F|.$$

Since  $p$  divides  $|G|$ ,  $p$  also divides  $|G| |\ker \phi|$ , and therefore  $p$  divides  $|F|$ .

(c) Since  $F$  is a direct product of the groups  $\mathbb{Z}_{k_i}$ , we have

$$|F| = k_1 \cdots k_n = \prod_{i=1}^n k_i.$$

$$\bigcup_{i=1}^{\infty} \sqrt{\phantom{x}}.$$

(d) Note that  $\langle g \rangle$  is cyclic with order  $m$ . By the fundamental theorem of cyclic groups, since  $p|m$ , there exists a subgroup  $\langle g^{m/p} \rangle$  with order  $p$ . It follows that  $|g^{m/p}| = p$ .

(e) If  $p$  is prime and  $p|ab$ , then either  $p|a$  or  $p|b$ . This can be generalized easily with induction to show if  $p|\prod_{i=1}^n a_i$  for  $a_i \in \mathbb{Z}$ , then there exists some  $i$  such that  $p|a_i$ . Therefore, since  $p$  divides the order of  $F$  by part (b), and since

$$|F| = \prod_{i=1}^n k_i$$

by part (c), there exists at least one  $k_i$  such that  $p|k_i$ . Since  $k_i$  is the order of  $a_i$ , there exists  $a_i \in G$  such that  $p||a_i|$ . By part (d), there exists an element of order  $p$  in  $G$ , namely  $a_i^{k_i/p}$ . ■

## Problem 2

(a) Because I don't feel like writing out a lot of subscripts and definitions, if I write  $\mathbf{m} \in \mathbb{Z}^n$ , then define  $\mathbf{m} = (m_1, \dots, m_n)$ . Additionally,  $a\mathbf{m} = (am_1, \dots, am_n)$ .

Let  $\phi : \mathbb{Z}^n \rightarrow \mathbb{Z}_8^n$  be the map  $g \mapsto g \pmod{8}$ . We will show  $\phi$  is a homomorphism. Let  $\mathbf{a}, \mathbf{b} \in \mathbb{Z}^n$ . It follows that

$$\begin{aligned}\phi(\mathbf{a} + \mathbf{b}) &= \phi(a_1 + b_1, \dots, a_n + b_n) \\ &= (a_1 + b_1, \dots, a_n + b_n) \pmod{8} \\ &\equiv (a_1, \dots, a_n) + (b_1, \dots, b_n) \pmod{8} \\ &= \phi(\mathbf{a}) + \phi(\mathbf{b}).\end{aligned}$$

Therefore,  $\phi$  is a homomorphism. We claim  $\ker \phi = a\mathbb{Z}^n$ . First, we will show  $a\mathbb{Z}^n \subseteq \ker \phi$ . Let  $\mathbf{z} \in a\mathbb{Z}^n$ . By definition, there exists  $\mathbf{v} \in \mathbb{Z}$  such that  $a\mathbf{v} = \mathbf{z}$ . Notice that for all components  $z_i$  of  $\mathbf{z}$ ,  $z_i = av_i$ . Since  $a|av_i$ ,  $z_i \equiv av_i \equiv 0 \pmod{a}$ , and thus  $\mathbf{z} \mapsto \mathbf{0}$ . Therefore,  $\mathbf{z} \in \ker \phi$ .

Now we show the converse. Let  $\mathbf{v} \in \ker \phi$ . It follows that  $a|v_i$  for all  $i$ , and thus there exist  $q_1, \dots, q_n \in \mathbb{Z}$  such that  $\mathbf{v} = (aq_1, \dots, aq_n) = a\mathbf{q}$ . Therefore,  $\mathbf{v} \in a\mathbb{Z}^n$  by definition. Therefore,  $\ker \phi = a\mathbb{Z}^n$ .

(b) Let  $\psi : G \rightarrow \text{Inn}(G)$  be the map  $g \mapsto \phi_g$ . We will show  $\psi$  is a homomorphism. Since  $\psi(ab) = \phi_{ab}$  and  $\psi(a)\psi(b) = \phi_a\phi_b$ , we must show  $\phi_{ab} = \phi_a\phi_b$ . Notice that

$$\phi_{ab}(x) = abx(ab)^{-1} = abxb^{-1}a^{-1} = \phi_a(bxb^{-1}) = \phi_a\phi_b(x).$$

Therefore,  $\psi$  is a homomorphism. Additionally, note that for any  $\phi_a \in \text{Inn}(G)$ , by definition  $a \in G$ , and thus  $\psi(a) = \phi_a$ . Therefore,  $\psi$  is surjective. We claim  $\ker \psi = Z(G)$ . Note that the identity in  $\text{Inn}(G)$  is  $\phi_e$ , and  $\phi_e(x) = x$ . First, we show  $Z(G) \subseteq \ker \psi$ . Let  $a \in Z(G)$ . It follows that  $ax = xa$  for all  $x \in G$ , and thus

$$\phi_a(x) = axa^{-1} = aa^{-1}x = x = \phi_e(x).$$

Therefore,  $a \mapsto \phi_e$ , and  $a \in \ker \psi$ . It follows that  $Z(G) \subseteq \ker \psi$ .

Now we show the converse. Let  $a \in \ker \psi$ . It follows that  $\phi_a = \phi_e$ , and thus  $axa^{-1} = x$  for all  $x \in G$ . Notice that

$$axa^{-1} = x \implies axa^{-1}a = xa \implies ax = xa.$$

Since this holds for all  $x \in G$ ,  $ax = xa$  for all  $x \in G$ , and thus  $a \in Z(G)$ . Therefore,  $\ker \psi \subseteq Z(G)$ , and thus  $\ker \psi = Z(G)$ .

## Problem 3

*Proof.* (a) Notice that  $HN = \{hn \mid h \in H \wedge n \in N\}$ . It follows that

$$HN = \bigcup_{h \in H} hN.$$

Since  $H \leq G$ , if  $h \in H$  then  $h \in G$ . Since  $N \triangleleft G$ , it follows that  $hN = Nh$  for all  $h \in H$ , and thus

$$HN = \bigcup_{h \in H} hN = \bigcup_{h \in H} Nh = NH.$$

Therefore,  $HN = NH$ , and  $HN$  is a subgroup of  $G$  by a previous homework problem. Let  $k \in HN$ . Since  $HN \leq G$ , it follows that  $k \in G$ . Since  $N$  is normal in  $G$ ,  $kN = Nk$  for all  $k \in HN$ . Therefore,  $N \triangleleft HN$ .

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(b) Let  $\phi : H \rightarrow HN/N$ , and let  $g, h \in H$ . Since  $N \triangleleft HN$  by part (a), it follows that

$$\phi(g)\phi(h) = gNhN = (gh)N = \phi(gh).$$

Therefore,  $\phi$  is a homomorphism.

(c) Let  $gN \in HN/N$ . Since  $g \in HN$ , there exist  $h \in H$  and  $n \in N$  such that  $g = hn$ . Since  $N = eN$  and  $n \in N$ , by properties of cosets  $nN = N$ , and thus  $hnN = h(nN) = hN$ . Since  $h \in H$ ,  $\phi(h) = hN$ , and thus  $\phi$  is surjective.

(d) First we show  $H \cap N \subseteq \ker \phi$ . Let  $g \in H \cap N$ . It follows that  $h \in N$ , and  $\phi(h) = hN$ . Since  $h \in N$ , by properties of cosets  $hN = N$ . Since the identity of  $HN/N$  is  $eN = N$ , we have  $H \cap N \subseteq \ker \phi$ .

Now we prove the converse. Let  $h \in \ker \phi$ . It follows that  $\phi(h) = hN = N$ . By properties of cosets,  $h \in N$ . Since  $\ker \phi \subseteq H$ ,  $h \in H$ . Therefore,  $h \in H \cap N$ . Therefore,  $\ker \phi = H \cap N$ .

(e) Since  $\phi$  is surjective by part (c),  $\phi(H) = HN/N$ . Since  $\ker \phi = H \cap N$  by part (d), and since  $\phi$  is a homomorphism by part (b), by the first isomorphism theorem it follows that

$$H/H \cap N = HN/N.$$

■

## Problem 4

(a) Let  $|G| = p^n$ . It follows from the class equation that

$$p^n = |Z(G)| + \sum_{a_i \notin Z(G)} [G : C(a_i)].$$

**Lemma 1.** *Let  $a, b, p \in \mathbb{Z}$  such that  $p|a$ . Then  $p|(a + b)$  if and only if  $p|b$ .*

*Proof.* Let  $a, b, p \in \mathbb{Z}$ , and let  $p|a$ . First, suppose  $p|b$ . It follows that there exist  $q, r \in \mathbb{Z}$  such that  $a = pq$  and  $b = pr$ . It follows that

$$a + b = pq + pr = p(q + r).$$

Since  $q + r \in \mathbb{Z}$ ,  $p|(a + b)$ .

Now suppose  $p|(a + b)$ . It follows that there exists  $q, r \in \mathbb{Z}$  such that  $a = pq$  and  $a + b = pr$ . It follows that

$$\begin{aligned} a + b &= pr \\ \implies pq + b &= pr \\ \implies b &= pr - pq = p(r - q) \\ \implies p &|b. \end{aligned}$$

Therefore, if  $p|a$ , then  $p|b$  if and only if  $p|(a + b)$ .

■

By Lagrange's theorem,  $|cl(a)| = |C(a)|$  divides  $p^n$ . It follows that  $[G : C(a)]$  divides  $p^n$ . Since the only divisors of  $p^n$  are powers of  $p$ , we know

$$\sum_{a_i \notin Z(G)} [G : C(a_i)]$$

is just a sum of powers of  $p$ , which we will write as

$$\sum_i p^{e_i}.$$

Notice that  $p|p^m$  for all integers  $m \neq 0$ . Notice that  $[G : C(a_i)] = 1$  implies  $C(a_i) = G$ , and thus  $a_i \in Z(G)$ . Therefore,  $p$  will divide  $[G : C(a_i)]$  for all  $a_i \notin Z(G)$ , and thus

$$p | \sum_i p^{e_i}.$$

Notice that  $p|p^n$ , and thus

$$p | \left( |Z(G)| + \sum_{a_i \notin Z(G)} [G : C(a_i)] \right).$$

By lemma 1, it follows that  $p$  divides  $|Z(G)|$ . Since  $e \in Z(G)$  implies  $|Z(G)| \geq 1$ , and since  $p \nmid 1$  for all  $p$ , we must have  $|Z(G)| > 1$ .

(b) By the fundamental theorem of finite abelian groups, it suffices to show  $G$  is abelian. Since  $|Z(G)| > 1$  and  $|Z(G)|$  divides  $p$ , we must have either  $|Z(G)| = p$  or  $|Z(G)| = p^2$ . For the first case, suppose for purpose of contradiction that  $|Z(G)| = p$ . Since  $Z(G) \triangleleft G$  and  $G$  is finite,

$$|G/Z(G)| = \frac{|G|}{|Z(G)|} = \frac{p^2}{p} = p.$$

By one of the corollaries of Lagrange's theorem,  $G/Z(G)$  is cyclic. By problem 5 in homework 7,  $G$  is abelian, and thus  $Z(G) = G$ , and thus  $|Z(G)| = p^2$ , which is a contradiction. Therefore,  $|Z(G)| = p^2$ .

Notice that if  $|Z(G)| = p^2 = |G|$ , since  $Z(G) \subseteq G$ , it follows that  $Z(G) = G$ , and thus  $G$  is abelian.

## Problem 5

**Definition 1.** Some definitions are necessary because this is going to get messy. Let  $\mathcal{L}$  be the set of left cosets of  $N(H)$  in  $G$ . Let  $\mathcal{C}$  be the set of subgroups conjugate to  $H$ . For all  $K \in \mathcal{C}$ , let  $\mathcal{G}(K)$  be the set of elements in  $G$  such that  $gHg^{-1} = K$ . Let  $\mathcal{D}$  be the set of all sets  $\mathcal{G}(K)$ . Symbolically, our definitions are

$$\begin{aligned}\mathcal{L} &= \{gN(H) \mid x \in G\} \\ \mathcal{C} &= \{gHg^{-1} \mid g \in G\} \\ \mathcal{G}(K) &= \{g \in G \mid gHg^{-1} = K\} \\ \mathcal{D} &= \{\mathcal{G}(K) \mid K \in \mathcal{C}\}.\end{aligned}$$

This proof is so messy that I think it's worthwhile to give a brief outline of the proof.

1. First, we show  $\mathcal{D} = \mathcal{L}$ . That is, the set of cosets is precisely the set of sets of elements that send  $H$  to a distinct conjugate.
2. Next, we define a bijection  $\mathcal{C} \rightarrow \mathcal{D}$  by  $K \mapsto \mathcal{G}(K)$ . It follows that  $|\mathcal{C}| = |\mathcal{D}| = |\mathcal{L}| = [G : N(H)]$ .

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First, we will show  $\mathcal{D} \subseteq \mathcal{L}$ . Let  $\mathcal{G}(K) \in \mathcal{D}$ . It follows that  $K \in \mathcal{C}$ . Fix some  $g \in \mathcal{G}(K)$ . We want to show  $\mathcal{G}(K) \in \mathcal{L}$ , namely  $\mathcal{G}(K) = gN(H)$ . First, we show  $\mathcal{G}(K) \subseteq gN(H)$ . Let  $x \in \mathcal{G}(K)$ . Notice that  $xHx^{-1} = K$ . Additionally, notice that for any  $gn \in gN(H)$ ,  $gnHn^{-1}g^{-1} = gHg^{-1} = K$ , since  $g \in \mathcal{G}(K)$  and  $n \in N(H)$ . It follows that

$$\begin{aligned} xHx^{-1} &= gHg^{-1} \\ \implies g^{-1}xHx^{-1}g &= H \\ \implies (g^{-1}x)H(g^{-1}x)^{-1} &= H. \end{aligned}$$

Therefore,  $g^{-1}x \in N(H)$ . By properties of cosets,  $gN(H) = xN(H)$ . By properties of cosets,  $x \in gN(H)$ . Therefore,  $\mathcal{G}(K) \subseteq gN(H)$ .

Now we show the converse. Let  $gn \in gN(H)$ . Notice that

$$gnH(gn)^{-1} = gnHn^{-1}g^{-1} = gHg^{-1} = K.$$

Therefore,  $gn \in \mathcal{G}(K)$ . Therefore,  $gN(H) = \mathcal{G}(K)$  for at least one fixed  $g \in \mathcal{G}(K)$ , and thus  $\mathcal{G}(K) \in \mathcal{L}$ . Therefore,  $\mathcal{D} \subseteq \mathcal{L}$ .

Now we show  $\mathcal{L} \subseteq \mathcal{D}$ . Let  $g \in G$  such that  $gHg^{-1} = K \in \mathcal{C}$ , where  $K$  is not necessarily a distinct subgroup from  $H$ . Let  $gN(H) \in \mathcal{L}$ . Notice that  $g \in \mathcal{G}(K)$ . We wish to show  $gN(H) = \mathcal{G}(K)$ . Let  $gn \in gN(H)$ . It follows that

$$gnH(gn)^{-1} = gnHn^{-1}g^{-1} = gHg^{-1} = K.$$

Therefore,  $gn \in \mathcal{G}(K)$ , and  $gN(H) \subseteq \mathcal{G}(K)$ . Now let  $x \in \mathcal{G}(K)$ . It follows that  $xHx^{-1} = gHg^{-1}$ , and thus

$$g^{-1}xHx^{-1}g = H.$$

Therefore,  $g^{-1}x \in N(H)$ , and by properties of cosets,  $gN(H) = xN(H)$ . By properties of cosets,  $x \in gN(H)$ , and  $\mathcal{G}(K) = gN(H)$ . Therefore,  $\mathcal{L} \subseteq \mathcal{D}$ .

Let  $\phi : \mathcal{C} \rightarrow \mathcal{D}$  be the map  $K \mapsto \mathcal{G}(K)$ . First, we show  $\phi$  is well defined. Let  $aHa^{-1} = bHb^{-1} = K \in \mathcal{C}$  with  $a \neq b$ . It follows that  $a, b \in \mathcal{G}(K)$ , and by properties of cosets,  $\mathcal{G}(K) = aN(H) = bN(H)$ . Therefore,  $\phi$  is well-defined.

Now we show  $\phi$  is surjective. Let  $\mathcal{G}(K) \in \mathcal{D}$ . By definition, it follows that  $K \in \mathcal{C}$ . Finally, we show  $\phi$  is injective. Let  $\mathcal{G}(K), \mathcal{G}(L) \in \mathcal{D}$  such that  $\mathcal{G}(K) = \mathcal{G}(L)$ . By definition, there exists  $a, b \in G$  such that  $K = aHa^{-1}$  and  $L = bHb^{-1}$ . By properties of cosets,  $a, b \in \mathcal{G}(K), \mathcal{G}(L)$ . Therefore,  $aHa^{-1} = K$  and  $aHa^{-1} = L$ , and thus  $K = L$ .

Therefore  $\phi$  is a bijection, and  $|\mathcal{C}| = |\mathcal{D}|$ . Since  $|\mathcal{D}| = |\mathcal{L}|$ , it follows that  $|\mathcal{C}| = |\mathcal{L}|$ . Since  $|\mathcal{L}| = [G : N(H)]$ , it follows that  $|\mathcal{C}| = [G : N(H)]$ .